

Answer to Q.1 Spectral decomposition of an Operator

Solution:

For the given matrix $\hat{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 4 \\ 1 & 0 & 2 \end{pmatrix}$:

Characteristic polynomial: $\det(\hat{A} - \lambda I) = -\lambda(3 - \lambda)^2 = 0$

Eigenvalues: $\lambda_1 = 0$ (multiplicity 1), $\lambda_2 = 3$ (multiplicity 2)

For $\lambda_1 = 0$: $|u_1\rangle = \frac{1}{\sqrt{61}} \begin{pmatrix} 6 \\ 4 \\ -3 \end{pmatrix}$ where $\|u_1\| = \sqrt{36 + 16 + 9} = \sqrt{61}$

For $\lambda_2 = 3$:

$$(\hat{A} - 3I) = \begin{pmatrix} -2 & 0 & 2 \\ 0 & 0 & 4 \\ 1 & 0 & -1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Only one independent eigenvector: $|u_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

Conclusion: Algebraic multiplicity (2) $\neq$ Geometric multiplicity (1) for $\lambda = 3$.

$\therefore$ Matrix is NOT diagonalizable. Spectral decomposition is not possible.

Diagonalizable example: Let $\hat{B} = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

Characteristic polynomial: $\det(\hat{B} - \lambda I) = (2 - \lambda)[(3 - \lambda)^2 - 1] = (2 - \lambda)(4 - \lambda)(2 - \lambda) = 0$

Eigenvalues: $\lambda_1 = 4$, $\lambda_2 = 2$, $\lambda_3 = 2$

For $\lambda_1 = 4$:

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow x = y, z = 0$$

$$|v_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

For $\lambda_2 = 2$ (multiplicity 2):

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow x + y = 0$$

Two independent eigenvectors: $|v_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$, $|v_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Spectral decomposition:

$$\hat{B} = 4|v_1\rangle\langle v_1| + 2|v_2\rangle\langle v_2| + 2|v_3\rangle\langle v_3|$$

Answer to Q.3 Bloch Sphere

Solution:

(a) $\hat{n} \cdot \vec{\sigma} = \sin \theta \cos \phi \sigma_x + \sin \theta \sin \phi \sigma_y + \cos \theta \sigma_z$

$$= \sin \theta \cos \phi \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \sin \theta \sin \phi \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \cos \theta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta & \sin \theta (\cos \phi - i \sin \phi) \\ \sin \theta (\cos \phi + i \sin \phi) & -\cos \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & -\cos \theta \end{pmatrix}$$

(b)

(i) $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$: $\theta = 0$, φ arbitrary

(ii) $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$: $\theta = \pi/2$, $\varphi = \pi/2$

(iii) $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: $\theta = \pi/2$, $\varphi = 0$

(c) $|\psi\rangle = \begin{pmatrix} \cos \theta/2 \\ e^{i\varphi} \sin \theta/2 \end{pmatrix}$

(i) $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$: $\cos(\theta/2) = \sin(\theta/2) = 1/\sqrt{2} \Rightarrow \theta = \pi/2$, $e^{i\varphi} = 1 \Rightarrow \varphi = 0$

(ii) $|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$: $\theta = \pi/2$, $e^{i\varphi} = -1 \Rightarrow \varphi = \pi$

(d)

(i) $\sigma_z|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \quad \checkmark$

$\sigma_z|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -|1\rangle \quad \checkmark$

(ii) $\sigma_x|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle \neq \lambda|0\rangle$

(iii) $\sigma_x|+\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle \quad \checkmark$

$\sigma_x|-\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -|-\rangle \quad \checkmark$

Answer to Q.6 General Unitary

Solution:

From Q.5(c), $R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$ and $R_y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$

$$\begin{aligned} R_y(\gamma)R_z(\delta) &= \begin{pmatrix} \cos(\gamma/2) & -\sin(\gamma/2) \\ \sin(\gamma/2) & \cos(\gamma/2) \end{pmatrix} \begin{pmatrix} e^{-i\delta/2} & 0 \\ 0 & e^{i\delta/2} \end{pmatrix} \\ &= \begin{pmatrix} e^{-i\delta/2} \cos(\gamma/2) & -e^{i\delta/2} \sin(\gamma/2) \\ e^{-i\delta/2} \sin(\gamma/2) & e^{i\delta/2} \cos(\gamma/2) \end{pmatrix} \end{aligned}$$

$$\begin{aligned} R_z(\beta)R_y(\gamma)R_z(\delta) &= \begin{pmatrix} e^{-i\beta/2} & 0 \\ 0 & e^{i\beta/2} \end{pmatrix} \begin{pmatrix} e^{-i\delta/2} \cos(\gamma/2) & -e^{i\delta/2} \sin(\gamma/2) \\ e^{-i\delta/2} \sin(\gamma/2) & e^{i\delta/2} \cos(\gamma/2) \end{pmatrix} \\ &= \begin{pmatrix} e^{-i(\beta+\delta)/2} \cos(\gamma/2) & -e^{-i(\beta-\delta)/2} \sin(\gamma/2) \\ e^{i(\beta-\delta)/2} \sin(\gamma/2) & e^{i(\beta+\delta)/2} \cos(\gamma/2) \end{pmatrix} \end{aligned}$$

Using $R_y(\theta) = e^{-i\theta Y/2}$ convention with $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ gives the matrix form shown.

Answer to Q.8 Quantum Gates – CNOT

Solution:

(a) Using CNOT $|ab\rangle = |a(a \oplus b)\rangle$:

$$\text{CNOT } |00\rangle = |0(0 \oplus 0)\rangle = |00\rangle$$

$$\text{CNOT } |01\rangle = |0(0 \oplus 1)\rangle = |01\rangle$$

$$\text{CNOT } |10\rangle = |1(1 \oplus 0)\rangle = |11\rangle$$

$$\text{CNOT } |11\rangle = |1(1 \oplus 1)\rangle = |10\rangle$$

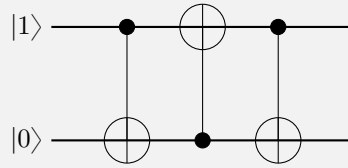
(b)(i) Starting with $|10\rangle$:

$$(\text{CNOT})_1 |10\rangle = |1(1 \oplus 0)\rangle = |11\rangle$$

$$(\text{CNOT})_2 |11\rangle = |1 \oplus 1, 1\rangle = |01\rangle \quad (\text{control on second qubit})$$

$$(\text{CNOT})_3 |01\rangle = |0(0 \oplus 1)\rangle = |01\rangle$$

(b)(ii) Quantum circuit:



(c) Computing each term in basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$\mathbb{I} \otimes \mathbb{I} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \sigma_z \otimes \mathbb{I} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\mathbb{I} \otimes \sigma_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \sigma_z \otimes \sigma_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

$$\begin{aligned} \text{CNOT} &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{aligned}$$

Answer to Q.9 Multiple CNOT

Solution:

Starting with general input $|ab\rangle$:

After first CNOT (control on top wire):

$$|ab\rangle \xrightarrow{\text{CNOT}_1} |a, (a \oplus b)\rangle = |\psi_1\rangle$$

After second CNOT (control on bottom wire):

$$|a, (a \oplus b)\rangle \xrightarrow{\text{CNOT}_2} |a \oplus (a \oplus b), (a \oplus b)\rangle = |b, (a \oplus b)\rangle = |\psi_2\rangle$$

Since $a \oplus (a \oplus b) = (a \oplus a) \oplus b = 0 \oplus b = b$

After third CNOT (control on top wire):

$$|b, (a \oplus b)\rangle \xrightarrow{\text{CNOT}_3} |b, b \oplus (a \oplus b)\rangle = |b, a\rangle$$

Similarly, $b \oplus (a \oplus b) = a$

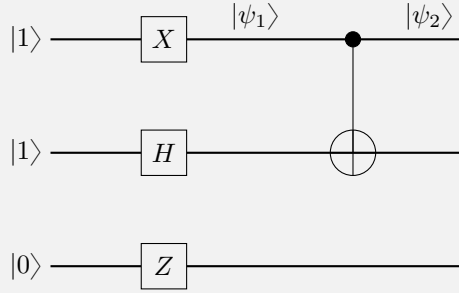
Therefore: $|ab\rangle \rightarrow |ba\rangle$ ✓

The circuit swaps the two qubits (SWAP gate).

Answer to Q.11 Quantum Circuits

Solution:

(a) Quantum circuit:



(b) $|\psi_1\rangle = (X \otimes H \otimes Z)|110\rangle$

$X|1\rangle = |0\rangle$, $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, $Z|0\rangle = |0\rangle$

$$\begin{aligned} |\psi_1\rangle &= |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |0\rangle \\ &= \frac{1}{\sqrt{2}}(|000\rangle - |010\rangle) \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \end{aligned}$$

(c) $|\psi_2\rangle = (\text{CNOT} \otimes I)|\psi_1\rangle$

CNOT acts on first two qubits:

$$\begin{aligned} |\psi_2\rangle &= (\text{CNOT} \otimes I) \frac{1}{\sqrt{2}}(|000\rangle - |010\rangle) \\ &= \frac{1}{\sqrt{2}}(\text{CNOT}|00\rangle \otimes |0\rangle - \text{CNOT}|01\rangle \otimes |0\rangle) \\ &= \frac{1}{\sqrt{2}}(|00\rangle \otimes |0\rangle - |01\rangle \otimes |0\rangle) \\ &= \frac{1}{\sqrt{2}}(|000\rangle - |010\rangle) = |\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \end{aligned}$$